

WEEK 1 REPORT

Requirement Analysis and Customer Database Module

1. Requirement Analysis

An e-commerce system is used to manage online buying and selling of products. I understood the main process of an online purchase and the data required to store customer and product information.

2. E-Commerce Process

The main steps in the e-commerce process are:

Browse / Search Product:

The customer searches for the product they want to buy.

Add to Cart:

After selecting the product, the customer adds it to the shopping cart.

Order Placement:

The customer selects the delivery address and places the order.

Checkout / Payment:

The customer proceeds to checkout and completes the payment using the available payment method.

Delivery:

After payment, the seller prepares and sends the product for delivery.

Review and Support:

After receiving the product, the customer can give a review. They can also ask for support if they need installation, return, or replacement.

3. Entities and Relationships

The main entities identified in the e-commerce database are:

ENTITY	DESCRIPTION	KEY ATTRIBUTES	RELATIONSHIP
Customer	Person who browses and purchases products	customer_id (PK), first_name, last_name, email, phone, created_at	One-to-Many with Order and Review

Seller	Merchants who lists products on the platform	seller_id (PK), company_name, email, phone, rating, bank_details	One-to-Many with Product
Product	Item available for sale on the platform	product_id (PK), seller_id (FK), title, description, price, stock_quantity, sku, category	Many-to-One with Seller, One-to-Many with Review
Order	Record of a purchase made by a customer	order_id (PK), customer_id (FK), order_date, status, total_amount, shipping_address	Many-to-One with Customer, One-to-One with Payment, Many-to-Many with Product through OrderItem
Payment	Transaction made for an order	payment_id (PK), order_id (FK), payment_method, status, transaction_id, amount, payment_date	One-to-One with Order
Review	Rating and feedback given by a customer for a product	review_id (PK), customer_id (FK), product_id (FK), rating, comment, created_at	Many-to-One with Customer and Product

Customer Table Creation:

```
Run SQL Command Line
SQL> CREATE USER E#COMMERCE IDENTIFIED BY Towel;
User created.
SQL> CREATE TABLE Customer (
2   customer_id INT PRIMARY KEY,
3   first_name VARCHAR(50),
4   last_name VARCHAR(50),
5   email VARCHAR(100),
6   phone VARCHAR(15),
7   created_at DATE
8 );
Table created.
SQL> desc customer;
Name                               Null?    Type
-----
CUSTOMER_ID                       NOT NULL NUMBER(38)
FIRST_NAME                        VARCHAR2(50)
LAST_NAME                         VARCHAR2(50)
EMAIL                            VARCHAR2(100)
PHONE                            VARCHAR2(15)
CREATED_AT                       DATE
SQL> commit;
Commit complete.
SQL> |
```

INSERT VALUES:

```
Run SQL Command Line x + -
SQL> commit;
Commit complete.

SQL> INSERT INTO Customer
  2 (customer_id, first_name, last_name, email, phone, created_at)
  3 VALUES (1, 'Rohana', 'Ranu', 'rohana@gmail.com', '9876543218', DATE '2026-09-01');
1 row created.

SQL>
SQL> INSERT INTO Customer
  2 (customer_id, first_name, last_name, email, phone, created_at)
  3 VALUES (2, 'Aisha', 'Khan', 'aisha@gmail.com', '9876543211', DATE '2026-09-02');
1 row created.

SQL>
SQL> INSERT INTO Customer
  2 (customer_id, first_name, last_name, email, phone, created_at)
  3 VALUES (3, 'Priya', 'Sharma', 'priya@gmail.com', '9876543212', DATE '2026-09-03');
1 row created.

SQL>
SQL> INSERT INTO Customer
  2 (customer_id, first_name, last_name, email, phone, created_at)
  3 VALUES (4, 'Ananya', 'Ravi', 'ananya@gmail.com', '9876543213', DATE '2026-09-04');
1 row created.

SQL>
SQL> INSERT INTO Customer
  2 (customer_id, first_name, last_name, email, phone, created_at)
  3 VALUES (5, 'Farah', 'Ali', 'farah@gmail.com', '9876543214', DATE '2026-09-05');
1 row created.

SQL>
SQL> INSERT INTO Customer
  2 (customer_id, first_name, last_name, email, phone, created_at)
  3 VALUES (6, 'Sneha', 'Ramu', 'sneha@gmail.com', '9876543215', DATE '2026-09-06');
1 row created.

SQL>
SQL> INSERT INTO Customer
  2 (customer_id, first_name, last_name, email, phone, created_at)
  3 VALUES (7, 'Rahul', 'Verma', 'rahul@gmail.com', '9876543216', DATE '2026-09-07');
1 row created.

SQL>
SQL> INSERT INTO Customer
  2 (customer_id, first_name, last_name, email, phone, created_at)
  3 VALUES (8, 'Meera', 'Nair', 'meera@gmail.com', '9876543217', DATE '2026-09-08');
1 row created.

SQL>
SQL> INSERT INTO Customer
  2 (customer_id, first_name, last_name, email, phone, created_at)
  3 VALUES (9, 'Arjun', 'Das', 'arjun@gmail.com', '9876543218', DATE '2026-09-09');
1 row created.

SQL>
SQL> INSERT INTO Customer
  2 (customer_id, first_name, last_name, email, phone, created_at)
  3 VALUES (10, 'Divya', 'Nolan', 'divya@gmail.com', '9876543219', DATE '2026-09-10');
1 row created.

SQL>
SQL> INSERT INTO Customer
  2 (customer_id, first_name, last_name, email, phone, created_at)
  3 VALUES (11, 'Anya', 'Roy', 'anya@gmail.com', '9876543220', DATE '2026-09-11');
```

CUSTOMER TABLE:

```

SQL> SELECT * FROM CUSTOMER;
CUSTOMER_ID FIRST_NAME
-----
LAST_NAME
-----
EMAIL
-----
PHONE CREATED_A
-----
1 Behana
Bamu
bchana@gmail.com
9876501234 01-SEP-26

CUSTOMER_ID FIRST_NAME
-----
LAST_NAME
-----
EMAIL
-----
PHONE CREATED_A
-----
2 Aisha
Rhan
alshad@gmail.com
9876543211 02-SEP-26

CUSTOMER_ID FIRST_NAME
-----
LAST_NAME
-----
EMAIL
-----
PHONE CREATED_A
-----
3 Priya
Shamra
priya@gmail.com
9876543212 03-SEP-26

CUSTOMER_ID FIRST_NAME
-----
LAST_NAME
-----
EMAIL
-----
PHONE CREATED_A
-----
4 Ananya
Soni
ananya@gmail.com
9876543213 04-SEP-26

CUSTOMER_ID FIRST_NAME
-----
LAST_NAME
-----
EMAIL
-----
PHONE CREATED_A
-----
5 Farah
Ali
afarah@gmail.com
9876543214 05-SEP-26

CUSTOMER_ID FIRST_NAME
-----
LAST_NAME
-----
EMAIL
-----
PHONE CREATED_A
-----
6 Sneha
Rumar
sneha@gmail.com
9876543215 06-SEP-26

CUSTOMER_ID FIRST_NAME
-----
LAST_NAME
-----
EMAIL
-----
PHONE CREATED_A
-----
7 Rahul
Verma
rahul@gmail.com

```

UPDATE METHOD:

```

SQL> UPDATE Customer SET phone = 9876501234 WHERE customer_id = 15;

1 row updated.

SQL> |

```

CONCLUSION:

I understood the basic requirements of an e-commerce database. I identified the main entities, their attributes, and relationships. I also learned the basics of creating the Customer table, inserting records, and updating existing data. This helped

me understand the initial structure of an e-commerce database system.